Regular Expressions

1. r"hi|hello|hey"

This matches:

- hi
- hello
- hey

Explanation:

- The | symbol means **"or"**.
- So it's saying: if the input is "hi" or "hello" or "hey", then use one of the associated responses.

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2. r"how are you?"

This matches:

- how are you?

Explanation:

- It's a simple **literal match**. If the user types exactly how are you? (including the question mark), it triggers this.

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3. r"(.*)color(.*)"

This matches:

- Anything that **contains the word "color"**, regardless of what's before or after it.

Examples it matches:

- "What's your favorite color?"
- "Tell me about color theory"

Explanation:

- (.*) means **"match anything (zero or more characters)"** before and after the word "color".
- So it's flexible to any sentence that includes the word "color".

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4. r"(.*)(course|program)(.*)"

This matches:

- Any input that **includes either "course" or "program"**.

Examples it matches:

- "Tell me about your course"
- "Do you have any program for developers?"

Explanation:

- (course|program) means it matches either "course" **or** "program".
- (.*) around it allows any text before or after.

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5. r"(.*)android(.*)"

This matches:

- Any sentence that contains the word **"android"**.

Examples:

- "Tell me about android"
- "Do you offer android training?"

Explanation:

- Again, (.*) lets it match any text surrounding the keyword "android".

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6. r"(.*)full stack(.*)"

This matches:

- Any sentence that contains the phrase **"full stack"**.

Examples:

- "What is full stack development?"
- "I'm interested in full stack"

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7. r"(.*)mobile\s?apps(.*)|(.*)applications(.*)"

This matches:

- Anything containing **"mobile apps"** (with or without space) OR the word **"applications"**.

Explanation:

- mobile\s?apps:
    - \s? = optional space between "mobile" and "apps"

    - So it matches "mobile apps" and "mobileapps"
- | = or
- (.*)applications(.*) matches any sentence containing "applications"

Examples:

- "Do you teach mobile apps?"
- "I want to build applications"

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8. r"(.*)"

This is the **default/fallback** pattern that matches **everything**.

Explanation:

- (.*) matches any string, so if no previous patterns match, this one always will.
- It's used to provide a generic response like "I didn't understand that".